



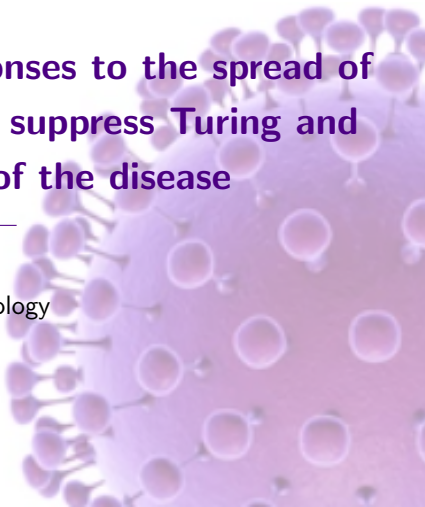
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Spatial behavioural responses to the spread of an infectious disease can suppress Turing and Turing–Hopf patterning of the disease

Giulia Gambini

Spatial models in Biology and Epidemiology

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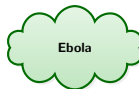


Introduction

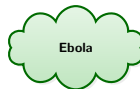


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- *The Reality of Adaptability:* Historical evidence shows that populations **actively adapt** their habits in response to perceived threats.
- *Behavioral Epidemiology (BE):* Investigates how humans can **adapt their behaviour to respond to the threats brought by infections**.





The study investigates a **family of reaction-diffusion models** for an **endemic SIR disease** under the following fundamental assumptions:

- 1 **No Pharmaceutical Intervention:** No vaccines or specialized treatments available.
- 2 **Implicit Awareness:** Behavioral adaptation is a direct function of local infection level (I) (no extra equations for news or media).

Two key **forms of behavioural responses**:

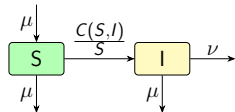
- **Social Distancing** (Local): Individuals reduce their contacts as infection grows.
- **Spatial Distancing** (Mobility): Individuals actively avoid visiting geographic sites where the risk is perceived as higher.



The Spatially Homogeneous Model

Reduced system

$$\begin{cases} \frac{dS}{dt} = \mu(1 - S) - C(S, I) \\ \frac{dI}{dt} = C(S, I) - (\mu + \nu)I \end{cases}$$



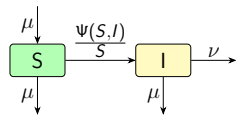
Where $C(S, I)$ is the total **infection rate**, a continuous function with properties:

- $C(S, 0) = C(0, S) = 0$: no infection occurs if there are no infectious subjects or no susceptibles.
- $\partial_I C(S, I) > 0$: The infection rate increases as the infectious prevalence grows.
- $\partial_S C(S, I) > 0$: The infection rate increases as the number of susceptible individuals grows.

Assumption: the population is able to enact **defensive measures**:

Reduced system with behavioural response

$$\begin{cases} \frac{dS}{dt} = \mu(1 - S) - \phi(I)C(S, I) \\ \frac{dI}{dt} = \phi(I)C(S, I) - (\mu + \nu)I \end{cases}$$



Where $\phi(S, I)$ is a suitable continuous and decreasing function of prevalence s.t. $0 \leq \phi(I) \leq 1$ and $\phi(0) = 1$.

$\Rightarrow C(S, I) =$ **Normal Incidence Rate (NIR)**

$\Psi(S, I) = \phi(I)C(S, I) =$ **Behavioural Incidence Rate (BIR).**



1. Disease-Free Equilibrium ($DFE = (1, 0)$)

$$\frac{dI}{dt} < 0 \iff \Psi(1 - I, I) < (\mu + \nu)I \quad \text{for } I > 0$$

$\Rightarrow$ The DFE is Globally Asymptotically Stable.

2. Endemic Equilibria ($EE = (S_e, I_e) > 0$)

- By setting the ODEs to zero, we can evaluate the endemic equilibria:

$$S = 1 - (1 + \frac{\nu}{\mu})I$$

$$\Psi(1 - (1 + \frac{\nu}{\mu})I, I) = (\nu + \mu)I$$

Due to the non-linearity of the BIR Ψ , the second equation can admit **zero, one, or multiple** equilibria.

3. Conditions for Local Stability of the EE:

$$\text{Tr}(J_{EE}^*) < 0 \implies 2\mu + \nu + \partial_S \Psi - \partial_I \Psi > 0 \quad (1)$$

$$\text{Det}(J_{EE}^*) > 0 \implies \mu(\mu + \nu - \partial_I \Psi) + (\mu + \nu)\partial_S \Psi > 0 \quad (2)$$



The Limit of General ODE Models

- Global behavior depends strictly on the specific functional form of $\Psi(S, I)$.
- Deep analytical investigation requires abandoning the meta-model to study **specific subcases**.

A General Exception: *Yakubovich's Theorem*

- **Condition:** The EE is unique and the DFE is unstable.
- **Implication:** The solutions of the system are strictly **oscillating**.

*To maintain a general meta-model approach, we now shift our focus from temporal ODEs to the onset of **spatial patterning**.*





1. Reaction-Diffusion Model

Introducing purely **random spatial mobility** (D_S, D_I), the system becomes:

Baseline Spatial system

$$\begin{cases} \partial_t S = D_S \Delta S + \mu(1 - S) - \Psi(S, I) \\ \partial_t I = D_I \Delta I + \Psi(S, I) - (\mu + \nu)I \\ \partial_n S|_{\partial\Omega} = \partial_n I|_{\partial\Omega} = 0 \end{cases}$$

2. Linear Stability Analysis

- * **Spatio-temporal perturbation** around the locally stable EE:

$$\begin{pmatrix} \delta S \\ \delta I \end{pmatrix} = \mathbf{c} e^{\lambda t} e^{i\mathbf{k} \cdot \mathbf{x}}$$

- * Linearizing the system and substituting the ansatz ($\Delta \rightarrow -k^2$) yields the **spatio-temporal Jacobian**:

$$\begin{aligned} J_{ST}(k) &= J_{EE} - \begin{pmatrix} D_S k^2 & 0 \\ 0 & D_I k^2 \end{pmatrix} \\ &= \begin{pmatrix} J_{11}^* - D_S k^2 & J_{12}^* \\ J_{21}^* & J_{22}^* - D_I k^2 \end{pmatrix} \end{aligned}$$



Characteristic polynomial of J_{ST} :

$$\lambda^2 + a_1(k)\lambda + a_0(k) = 0, \quad a_1(k) = -\text{Tr}(J_{ST}), \quad a_0(k) = \text{Det}(J_{ST})$$

In order to obtain **Turing Instability**, we require:

$$-\text{Tr}(J_{ST}) = a_1(k) = (D_S + D_I)k^2 - (\text{Tr}(J_{EE})) > 0$$

$$\text{Det}(J_{ST}) = a_0(k) = (D_S D_I)k^4 - (D_I J_{11}^* + D_S J_{22}^*)k^2 + \text{Det}(J_{EE}) < 0$$

$$\Rightarrow D_I J_{11}^* + D_S J_{22}^* > 0$$

We also need require to have $\Delta > 0$, which yields

$$D_I J_{11}^* + D_S J_{22}^* > 2\sqrt{D_S D_I \text{Det}(J_{EE})}$$

In conclusion, by substituting $J_{11}^* = -\mu - \partial_s \Psi$ and $J_{22}^* = \partial_I \Psi - (\mu + \nu)$ we obtain the **Conditions for Turing Instability**:

$$\text{Tr}(J_{EE}) < 0 \quad (3)$$

$$\text{Det}(J_{EE}) > 0 \quad (4)$$

$$D_S(\partial_I \Psi - (\mu + \nu)) - D_I(\mu + \partial_S \Psi) > 2\sqrt{D_S D_I \text{Det}(J_{EE})} \quad (5)$$

Results:

- * r.h.s. is positive $\Rightarrow$ if $\partial_I \Psi - (\mu + \nu) \leq 0$, no Turing patterns.
- * From $\text{Tr}(J_{EE}) < 0$ and by multiplying by $(1 - F^2) < 1$, we obtain $\partial_I \Psi - (\mu + \nu) = (\mu + \partial_S \Psi)(1 - F^2)$.

Substituting into the condition (5):

$$D_S > \frac{D_I}{1 - F^2} + \alpha_0^2 > D_I, \quad \text{where} \quad \alpha_0^2 = \frac{2\sqrt{D_S D_I a_0(0)}}{(\mu + \partial_S \Psi)(1 - F^2)} \geq 0$$

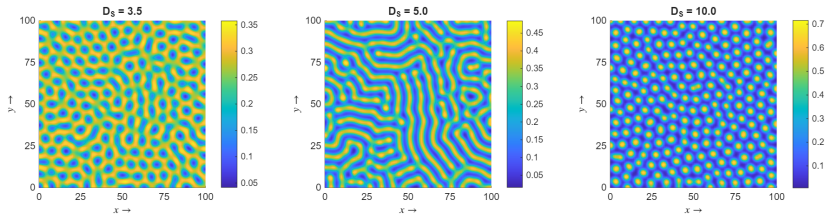
Conclusion: A necessary condition for Turing patterns is $D_S > D_I$.

This particular model does not depend on human behaviour, so that $\phi(I) = 1$, $C(S, I) = \beta SI^2$, then $\Psi(S, I) = \beta SI^2$. Which yields:

The Sun model

$$\begin{cases} \partial_t S = D_S \Delta S + \mu(1 - S) - \beta SI^2 \\ \partial_t I = D_I \Delta I + \beta SI^2 - (\mu + \nu)I \end{cases}$$

Setting $\beta = 35$, $\mu = 1$ and $\nu = 1.8$:



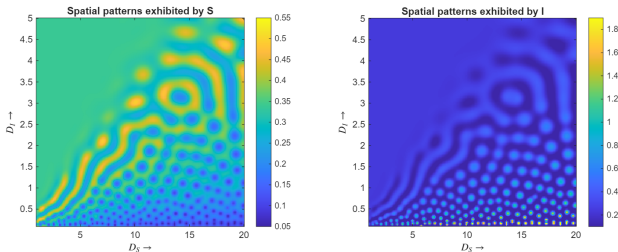
Turing patterns of I population depending on D_S for $D_I = 1$. Left panel: $D_S = 3.5$; Central panel: $D_S = 5$; Right panel: $D_S = 10$.

Simulation Parameters:

- **Domain & Grid:** $\Omega = [0, 200] \times [0, 200]$, with $\Delta x = \Delta y = 0.5$ and $\Delta t = 0.001$, and $t_{end} = 2000$.
- **Initial Conditions:** perturbed Endemic Equilibrium

$$(S_0, I_0) = (S_{EE}(1 + \rho_S), I_{EE}(1 + \rho_I))$$

where $\rho_S(x, y), \rho_I(x, y) \in (-0.01, 0.01)$



Visual summary of spatial Turing patterns exhibited by S (panel (a)) and I (panel (b)) for a range of values of D_S and D_I .



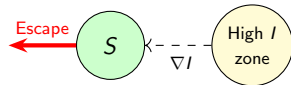
Now we introduce a **spatial behavioural response**, modeled by adding to the spatial flow of S subjects a new component: $\psi = -D\nabla S - \bar{A}S\nabla I$

Baseline Spatial system

$$\begin{cases} \partial_t S = D_S \Delta S + \operatorname{div}(\bar{A}S\nabla I) + \mu(1 - S) - \Psi(S, I) \\ \partial_t I = D_I \Delta I + \Psi(S, I) - (\mu + \nu)I \\ \partial_n S|_{\partial\Omega} = \partial_n I|_{\partial\Omega} = 0 \end{cases}$$

* $\operatorname{div}(\bar{A}S\nabla I)$: **non-linear cross-diffusion term**, it mathematically represents *chemorepulsion*.

* $\bar{A} > 0$: **strength of the spatial distancing**.



As before, perturbing and linearizing around the *locally stable homogeneous steady state* (S_{EE}, I_{EE}):

$$J_{ST}(k) = \begin{pmatrix} J_{11}^* - D_S k^2 & J_{12}^* - Ak^2 \\ J_{21}^* & J_{22}^* - D_I k^2 \end{pmatrix}, \quad \text{where } A = \bar{A}S_{EE}$$



The addition of the Ak^2 term modifies the zero-degree coefficient of the characteristic polynomial: $b_0(k) = a_0(k) + J_{21}^* Ak^2$.

Since $J_{21}^* > 0$, **the spatial escape exerts an intrinsically stabilizing effect.**

By setting $b_0(k) = 0$ we can find the critical threshold required for keeping the determinant positive forever:

The suppression condition

$$A_{cr} := \frac{D_I J_{11}^* - 2\sqrt{D_s D_I b_0(0)}}{J_{21}^*}$$

- * for $A > A_{cr}$: the EE is LAS, i.e. if the behavioural spatial response is sufficiently large, then the Turing Instability pattern disappears.
- * for $A < A_{cr}$: there exists an interval of frequencies $(k_1(A), k_2(A))$ in which the system is unstable.



The system is unstable only within the frequency band $(k_1(A), k_2(A))$.

To understand the pattern evolution, we isolate the **distancing parameter**

$$B = \frac{J_{21}^* A}{D_S D_I}$$

The derivatives of the roots of the characteristic polynomial $k_1^2(B)$ and $k_2^2(B)$ reveal a crucial property:

$\partial_B k_1^2(B) > 0 \implies$ The lower bound **increases**.

$\partial_B k_2^2(B) < 0 \implies$ The upper bound **decreases**.

Conclusions

- As spatial distancing (A) increases towards A_{cr} , the instability interval **shrinks**.
- As a consequence, this proves that varying $A \in (0, A_{cr})$ induces a pattern transition (e.g., from labyrinths to spots) observed in the numerical simulations.

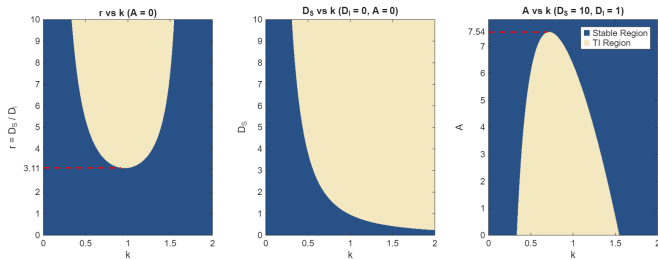
The Sun model with spatial distancing



We modify the Sun model by adding the spatial distancing term $\text{div}(\bar{A}S\nabla I)$, yielding the following modified system:

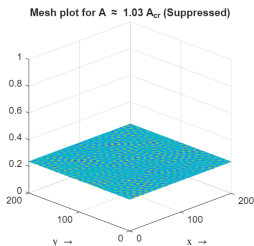
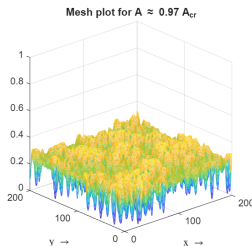
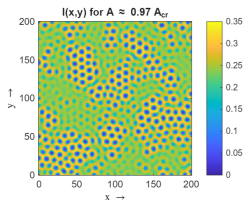
The Sun model with spatial distancing

$$\begin{cases} \partial_t S = D_S \Delta S + \mu(1 - S) + \text{div}(\bar{A}S\nabla I) - \beta SI^2 \\ \partial_t I = D_I \Delta I + \beta SI^2 - (\mu + \nu)I \end{cases}$$



Impact of (D_S, D_I) on Turing Instability vs Local Stability for $A = 0$ left and central panels, $A > 0$ right panel. $R = D_S/D_I$, spatial frequency, $\beta = 35$, $\mu = 1$ and $\nu = 1.8$.

- * **Sub-critical distancing** ($A < A_{cr}$): Insufficient spatial distancing allows the instability to grow, leading to the formation of high-prevalence epidemic clusters.
- * **Super-critical distancing** ($A > A_{cr}$): The spatial distancing acts as a shield. It suppresses the instability, flattening the epidemic wave to a uniform steady state.

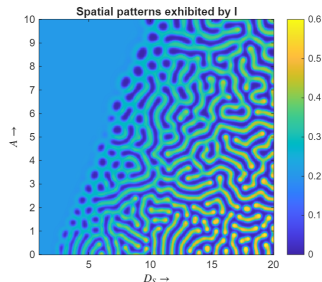
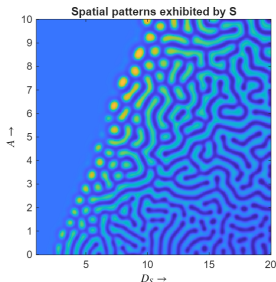


Impact of spatial-distancing parameter A on the steady state I . Left and central panel: density and mesh plots of $I(t_f, x, y)$ for $A \approx 0.97 A_{cr}$. Right panel: mesh plot of $I(t_f, x, y)$ for $A \approx 1.03 A_{cr}$.

To simulate the behavioural response, we assume that social alertness (q) reduces local contacts, setting $\phi(I) = \frac{1}{1+qI^2}$.

The Sun model with spatial distancing

$$\begin{cases} \partial_t S = D_S \Delta S + \mu(1 - S) - \frac{\beta SI^2}{1+qI^2} + \operatorname{div}(\bar{A} S \nabla I) \\ \partial_t I = D_I \Delta I + \frac{\beta SI^2}{1+qI^2} - (\mu + \nu)I \end{cases}$$



Spatial patterns for S (left) and I (right) varying D_S and A . Fixed parameters: $\beta = 35$, $\mu = 1$, $\nu = 1.8$, $q = 2$, $D_I = 1$.



Spatial Distancing annihilation of Hopf-Turing Instability



*What happens if the instability of the Endemic Equilibrium arises through a **Hopf bifurcation**?*

- Near the bifurcation, the determinant remains positive ($a_0(k) > 0$).
- Characteristic polynomial: $\lambda^2 - 2z\omega_0\lambda + \omega_0^2$,
 $\omega_0 > 0$ is the oscillation pulsation, $z \in (0, 1)$ the damping coefficient.

Stability: $a_1(k) = (D_S + D_I)k^2 - 2z\omega_0$

Symmetry-breaking scenarios:

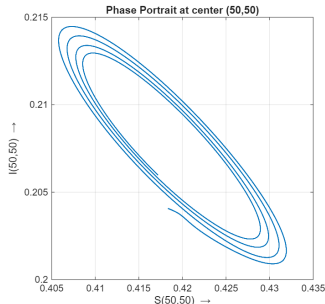
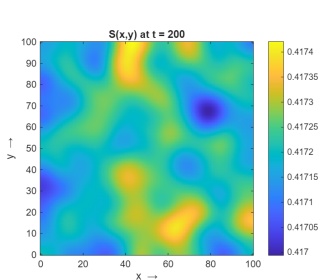
* **Spatio-temporal Hopf instability:**

$$a_0(k) > 0 \quad \forall k \text{ but } a_1(k) < 0 \text{ for } 0 \leq k^2 < k_H^2 = \frac{2z\omega_0}{D_S + D_I} \ll 1.$$

Result: Oscillations in time, but uniformity in space.

- * **Turing-Hopf Instability:** $a_0(k) < 0$ and $a_1(k) < 0$ for some k .

Result: The system enters the Turing-Hopf domain: **simultaneous breaking of both temporal and spatial symmetries.**



Spatio-temporal oscillations for $D_I = 0$, $D_S = 2$, $A = 7.66$, $q = 2$. Left: $S(x, y, 200)$; Right: Phase portrait.

Spatio-temporal chaos: Localized hotspots "pulse" and evolve over time, creating unpredictable epidemic fluctuations.





- **Beyond passive models:** Human behavioral mechanisms profoundly manipulate the spatial structure of epidemic outbreaks.
- **The power of Spatial Distancing (A):** Cross-diffusion acts as an active, non-local stabilizer.
 - Sub-critical ($A < A_{cr}$): Induces pattern transitions, altering the shape of epidemic clusters (e.g., from labyrinths to spots).
 - Super-critical ($A > A_{cr}$): Completely annihilates complex pattern formation and restores the homogeneous steady state.
- **Turing-Hopf Instability:** Purely local responses fail to eradicate spatio-temporal chaos. However, it can be proven numerically that spatial distancing (A) plays a key role in extinguishing unpredictable epidemic waves.